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A hybrid population-based algorithm for solving the fuzzy capacitated maximal covering location problem

Méziane Aider [a,1], Imene Dey [a,1], Mhand Hifi [b,1,*]

[a] USTHB, LaROMaD, BP 32 El Alia 16111 Bab Ezzouar, Alger, Algeria
[b] UPJV, EPROAD, UR 4669, 7 rue du Moulin Neuf, 80000 Amiens, France

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ABSTRACT

In this paper, we study the fuzzy capacitated maximal covering location problem, where a hybrid population-based algorithm is proposed to solve it. An instance of the problem is represented by a set of customers in a given network with their distances and such that the coverage degree of facilities and the distance between customers are fuzzy. Its goal is to open (select) a subset of facilities to be located on customers such that the coverage of customers should be maximized. Two versions of a hybrid population-based algorithm are designed for tackling the aforementioned problem: (i) a modified grey wolf optimizer combined with exploration and exploitation strategies, and (ii) a modified grey wolf optimizer combined with a series of local searches and an upper bound delimiting wolf movements. Each version of the method enhances the quality of the solutions belonging to the current population by adding a series of local searches. Further, the exploration strategy is based upon the drop/rebuild operator for jumping from the current region to unvisited ones. Finally, the behavior of the proposed method is evaluated on a set of benchmark instances of the literature, where its provided results are compared to those reached by more recent methods of the literature and the state-of-the-art Cplex solver. Encouraging results have been obtained.

1. Introduction

The Maximal Covering Location Problem (namely MCLP) is an NP-hard combinatorial optimization problem, which arises in several real-world applications like in searching for the optimal location of emergency response facilities, services, and vehicles (Li et al., 2011), communication networks (Lee & Murray, 2010), security monitoring sensors (Murray et al., 2007) and, retail stores (Frank & Lieselot, 2007).

The well-known MCLP is based on finding the optimal location of a subset of given facilities in order to cover the maximal number of customers positioned on a network, where each facility's coverage area and the distance between a customer and a facility, are fixed. Such constraints are not always available in reality. As an example, one can observe the constraints considered for the mobile phones network and the web-internet networks (Davari et al., 2011). Generally, a straightforward version of that problem, which also be considered as an extended MCLP with fuzzy distances, was studied and called the Fuzzy Capacitated Maximal Covering Location Problem (namely FCMCLP). The goal of FCMCLP is to establish the best opened (selected) locations regarding a fixed number of facilities that covers a maximum number of customers in a given network, such that both the distance between the customers and the installed facilities, and the coverage area of each facility are fuzzy.

1.1. An instance of the studied problem

In this paper, we study FCMCLP, where its instance is defined by a set P of m points (positions) in a plane and a set of k, $1 \leq k \leq m$, facilities to be installed at a chosen subset of points. Each customer $p_i \in P$, $i = 1, \ldots, m$, is represented by a non-negative demand w_i while each located facility is represented by a supply capacity s_j, $1 \leq j \leq m$. A facility can be positioned within its coverage area center c_j, which is considered as two concentric circles, with R radius and $R + f_r$ fuzzy radius, respectively. The centers of the aforementioned circles are restricted to the locations of customers themselves (Lorena & Pereira, 2002) while the coverage degree of a facility c_j positioned at a point p_i, based on the exact definition of Drakulic et al. (2016), is given as follows:

$$Cov(i,j) = \begin{cases} 1 & \text{if } d(p_i, c_j) \leq R \\ \frac{1}{2}(\mu(x_1) + \mu(x_2)) & R < d(p_i, c_j) \leq R + f_r \\ 0 & d(p_i, c_j) > R + f_r, \end{cases} \tag{1}$$

* Corresponding author.
E-mail addresses: meziane.aider@gmail.com (M. Aider), idey@usthb.dz (I. Dey), hifi@u-picardie.fr (M. Hifi).

[1] All authors are listed in alphabetical order of their last name, according to the Operational Research domain in France.

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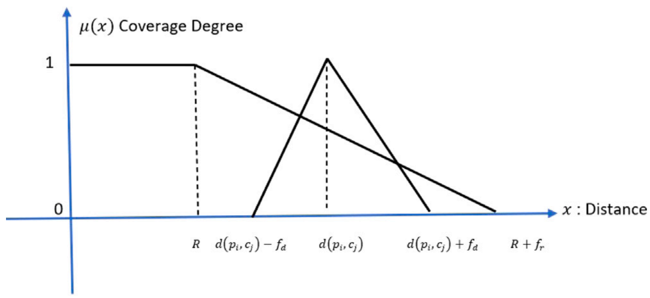

Fig. 1. Illustration of the standard degree of coverage.

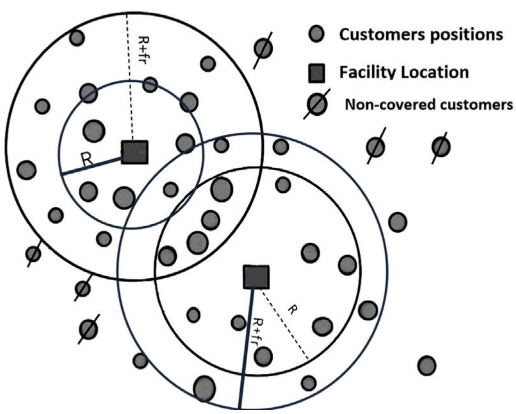

Fig. 2. Illustration of the coverage of two opened facilities for a given instance.

where, on the one hand, $d(p_i, c_j)$ denotes the Euclidean distance between the positioned facility c_j and the customer p_i, and on the other hand, $d(p_i, c_j) \pm f_d$ denotes its fuzzy distance, such that f_d denotes a real number belonging to the interval $[5, 100]$ (Drakulic et al., 2016). Fig. 1 illustrates both values $\mu(x_1)$ and $\mu(x_2)$ used by the formula 1.1, where the coverage degree may be computed for each used instance (as considered in Drakulic et al., 2016).

Fig. 2 shows a set of customers represented on the plane. On the one hand, one can observe that two facilities were positioned for covering the maximum number of overall customers. On the other hand, the same figure illustrates the two centering circles related to a given located facility, including both the radius R and the fuzzy radius $R + f_r$.

1.2. Contribution

The design of powerful methods, for various combinatorial optimization problems, remains a challenge for searching solutions with high qualities. Because searching for optimal solutions remains intractable, especially when tackling "large-sized" or "arduous" instances, it is often paramount for searching good "sub-optimal" solutions for practical situations. Population-based methods, like scatter search, swarm optimization and others, have shown some advantages when adapted to different problems of this category.

In recent years, it has been observed that the hybridization of solution procedures becomes a very promising research issue. In fact, it was observed that diversified algorithms have advantages when combining them with exploration and exploitation operators. We believe that their cooperation induces a good balance between multiple-average solutions and those with high quality. Further, with the emergence of the grey wolf optimization method, several studies have been conducted to address various discrete and continuous optimization problems. Each of the studies differs, even if the goal is to design a "good" experimental method able to compete with other methods available in the literature.

Herein, we design a powerful hybrid population-based algorithm for solving FCMCLP. The designed method starts with a reference set of solutions (related to the wolves) and expands it by considering both exploration and exploitation operators. There are two versions of the method: (i) the first version adopts the principle of the discrete grey wolf optimizer hybridized with intensification and restarting search, and (ii) the second version applies the same principle as the first one except that the generation of the reference set is done around a given upper bound and the positions associated to each solution are the result of the combination between the best solutions of the reference set that is delimited by this upper bound.

The principle of the proposed method is based on the following features:

1. To build a *reference set* used as a starting population for the designed method: A first initial solution is obtained by using an iterative generator and the expanded set of solutions is built around this initial solution (ensuring the feasibility of the solutions). Moreover, the first version of the method uses a random generator for providing the starting reference set, while the second version focuses on a generation around a given upper bound.
2. Then, the following two stages are repeated:
   (a) Apply the grey wolf optimizer-based procedure for enriching the pack composed of the three best solutions and the rest of the population.
   (b) Apply a drop and rebuild operator whenever the population stagnates. In this case, the exploration of new sub-spaces is considered; which is used for diversifying the search process.
3. Steps 2a and 2b are iterated till matching the final stopping condition.

The rest of the paper is organized as follows. Section 2 reviews some works related to the location problem with some of its variants. A formal description of FCMCLP is presented in Section 3.1. In Section 3, we first discuss the basic principle of the grey wolf optimizer, and next, we present the hybrid population-based algorithm designed for approximately solving FCMCLP. In the same section, an enhanced version of the algorithm is presented. Finally, in Section 4, the behavior of the designed algorithm is tested and analyzed on instances taken from the literature, where its achieved results are compared to those provided by a recent method of the literature and the state-of-the-art Cplex solver.

2. Background

MCLP is a classical covering location problem, first introduced by Church and ReVelle (1974). Such a problem aims in searching for the best combination of p facilities that is able to maximize the covered demand of m customers. MCLP, with its variants, may be encountered in a variety of fields, like police patrol planning (Nodarse et al., 2019), location of Wi-Fi antennas (Lee & Murray, 2010), distribution of drones for medical care (Pulver et al., 2016) and, location of mining bases (Xue et al., 2017).

Due to the complexity of the studied problem (with its variants), there are several available papers tackling it in the literature. Among the first authors who tackled the basic version of MCLP, we can cite (Galvao & ReVelle, 1996) who designed a Lagrangian heuristic. Such an approach used the sub-gradient optimization to maximize the Lagrangian dual hoping to provide a series of tight upper and lower bounds. The experimental part analyzed the behavior of the method and compared its performance to another Lagrangian-based approach on randomly generated benchmark instances containing 150 facilities.

Davari et al. (2013) tackled large-scale MCLP by combining variable neighborhood search and fuzzy simulation. Test problems with up to 2500 facilities showed that their method was very competitive since it was able to find solutions with gaps less than 1.5% in a reasonable run-time when compared its performance to exact methods.

For the same problem, Atta et al. (2018) proposed a mimetic algorithm, in which the genetic algorithm was combined with a special refinement operator while Zarandi et al. (2011) proposed another customized genetic algorithm to solve large-scale MCLP instances, with up to 2500 facilities.

For the Capacitated MCLP (noted CMCLP), where the finite capacity constraint of each facility has been imposed, Pirkul and Schilling (1991) tackled it by proposing a formal description based on the strong form of the capacitated p-median problem.

Next, a Lagrangian relaxation-based solution procedure was designed for approximately solving that problem. The experimental part underlined the performance of such a method, where it was able to solve randomly generated instances containing between 50 and 100 facilities up to thirty tasks within two minutes.

A particle swarm optimization was developed by ElKady and Abdelsalam (2016) for approximately solving CMCLP. In the experimental part, the author compared the results achieved by the method to those reached by the state-of-the-art Cplex solver; the authors underlined the effectiveness of the method in determining good locations for the facility, maximizing demand coverage.

In Davari et al. (2013) a variable neighborhood search-based solution procedure was proposed for tackling another variant; that is the MCLP with fuzzy coverage radii for facilities. In their computational part, the authors pinpointed the good behavior of the method, especially on instances with up to 2500 nodes. It was also able to obtain lower bounds (feasible solutions) with a gap smaller than or equal to 1.5% in much less average runtime when compared to accurate algorithms.

Whenever both the coverage radii of facilities and the distance between a customer and, a facility are considered fuzzy, Drakulic et al. (2016) designed a special swarm optimization for tackling it. The computational results contains two parts: (i) a first part including a compared study between their method and the state-of-the-art Cplex solver, especially on small-sized instances containing between 70 and 90 locations, and (ii) a second part including large-scale instances, containing between 100 and 900 locations, in which the behavior of the method was studied.

Davari et al. (2011) combined the fuzzy simulation and simulated annealing for tackling the Fuzzy MCLP (noted FMCLP). In their experimental part, the authors studied the behavior of the method in some instances, where its performance was analyzed and affirmed that the global average objective values are less than or equal to 1.35% of the overall average optimal values.

For an extension of the fuzzy maximal covering location time problem, where travel times may be linear, trapezoidal, or Gaussian fuzzy numbers, Takaci et al. (2012) proposed an adaptation of the particle swarm optimization to solve it. In the experimental part, the authors demonstrated the effectiveness of their method when comparing its results to those published in the literature.

Atta et al. (2022) proposed a multiple-metaheuristic for solving the fuzzy capacitated maximal covering location problem. The particle swarm optimization, differential evolution and two generation metaheuristics (artificial bee colony algorithm and firefly algorithm) were adapted for approximately solving the problem. In their experimental part, an extensive comparative study was conducted on 80 benchmark instances including medium-sized and large-scale instances. Further, the results provided by all considered methods were compared, and the best achieved results were compared to those reached by the Cplex.

We note that there are other papers, where nice resolution methods have been designed for several variants of the covering location (Farahani et al., 2012; Owen & Daskin, 1998) and, for more complete reviews on the domain, the reader can be referred to Snyder and Haight (2016) and Drezner and Hamacher (2002).

3. A hybrid population-based algorithm for FCMCLP

In this section, we first express the mathematical formulation of FCMCLP, already considered in the literature (Atta et al., 2022). Next, we discuss the linear relaxation of the aforementioned model for establishing an upper bound for the hybrid method. Further, we describe the basic principle of the grey wolf optimizer, including its main steps. Finally, we discuss how such a method can be adapted (and enhanced) for tackling the FCMCLP.

3.1. FCMCLP

Let x_{ij} be a decision variable related to linking the position p_i of the customer i and the facility c_j such that :

$$x_{ij} = \begin{cases} 1 & \text{if customer } i \text{ is served by facility } j \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

and y_j be a decision variable related to a facility j such that

$$y_j = \begin{cases} 1 & \text{if a facility is opened at the location j} \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

Let $I = J = \{1, \dots, m\}$; then, a formal description of FCMCLP may be stated as follows:

$$\max \; z(x) = \sum_{i=1}^{m} \sum_{j=1}^{m} (w_i \times cov_{ij}) x_{ij} \tag{4}$$

$$\text{s.t.} \quad \sum_{j=1}^{m} y_j = k \tag{5}$$

$$\sum_{j=1}^{m} x_{ij} \leq 1, \forall i \in I \tag{6}$$

$$\sum_{i=1}^{m} (w_i \times cov_{ij}) x_{ij} \leq s_j y_j, \forall j \in J \tag{7}$$

$$x_{ij} \in \{0,1\}, y_j \in \{0,1\}, \; i \in I, j \in J, \tag{8}$$

where equation (4) denotes the objective function that maximizes the sum of demands satisfied by the opened (positioned/selected) facilities according to their coverage degrees. The constraints (5) make sure that only k facilities have been positioned on the network. Constraints (6) verify if each customer's demand is satisfied by at most one opened facility. Each inequality (7) requires that the global demand satisfied by the facility j is at most equal to the total capacity of the facility j, which explains that a given facility j can partially cover a given customer i with a certain coverage degree cov_{ij}, and the sum of the whole demands of the partially(or completely) covered customers by this facility are within its capacity s_j. Finally, the last line (9) of the model is related to the domain of the decision variables.

We have already mentioned the need of an upper bound in order to make the solutions of the referent set cooperate with this bound. Let $\text{RP}_{\text{FCMCLP}}$ be the resulting program induced from FCMCLP by relaxing overall binary variables, i.e., setting $x_{ij} \in [0,1]$, and $y_j \in [0,1], \forall i \in I, \forall j \in J$ such that $\text{RP}_{\text{FCMCLP}}$ is given as follows:

$$\max \; z(x) = \sum_{i=1}^{m} \sum_{j=1}^{m} (w_i \times cov_{ij}) x_{ij}$$

$$\text{s.t.} \quad \sum_{j=1}^{m} y_j = k$$

$$\sum_{j=1}^{m} x_{ij} \leq 1, \forall i \in I$$

$$\sum_{i=1}^{m} (w_i \times cov_{ij}) x_{ij} \leq s_j y_j, \forall j \in J$$

$$x_{ij} \in [0,1], y_j \in [0,1], \; i \in I, j \in J, \tag{9}$$

The relaxation RP_{FCMCLP} is optimized by using the Simplex method. Then, we collect the current primal solution representing the primal configuration; that is an upper bound for the original problem FCMCLP. Of course, one can observe that relaxing only the decision variables x_{ij} may provide a tighter upper bound. However, limited computational results showed that relaxing all variables provided an upper bound very close to the latter, but required much more runtime, especially for large-sized instances. Hence, we limit this study to the bound produced by the Simplex method.

3.2. A basic grey wolf optimizer

The Grey Wolf Optimizer (GWO) is a recent meta-heuristic inspired from the grey wolves in nature. Often, such an approach tries to mimic the leadership hierarchy and hunting mechanism of grey wolves, and to use it as a search process for generating series of positions related to feasible solutions to a given problem. The grey wolf lives in groups limited to a small number of wolves, which contains the leader wolf, noted α, who drives the hunting. A standard hierarchy leadership has the wolf, noted β, on its second level, which is the next best wolf in the pack, and then it follows the third level known as the wolf δ. The lowest ranking grey wolves represent a subgroup of the rest of the wolves, noted ω each, where each of them respects the decisions provided by the three dominant wolves, i.e., α, β and δ.

In addition to the social hierarchy of wolves, group hunting is another interesting social behavior of grey wolves. According to Muro et al. (2011), the main phases of grey wolf hunting are represented by tracking, encircling, and attacking the prey. Formulas related to the social hierarchy with the main phases of hunting are given in what follows, such that the wolves use a predefined direction to encircle the prey, where each wolf's position is updated according to the prey's position:

$$\vec{D} = |\vec{C} \cdot \vec{S_p}(t) - \vec{S}(t)| \tag{10}$$

where the new position related to each wolf will be updated as follows:

$$\vec{S}(t+1) = \vec{S_p} - \vec{A} \cdot \vec{D}, \tag{11}$$

which aims to encircle the prey. In this case, for Eq. (11), t denotes the current iteration, $\vec{S}(t+1)$ is the position vector of the wolf, $\vec{S_p}$ denotes the position vector of the prey, while $\vec{A}$ and $\vec{C}$ are two coefficient vectors defined as follows:

$$\vec{A} = 2\vec{a} \cdot \vec{r_1} - \vec{a} \quad \text{and} \quad \vec{C} = 2\vec{r_2} \tag{12}$$

On the one hand, $\vec{a}$ (used in the first equation of (12), on the left-hand) is a vector initialized to 2 and decreases till matching the value 0 throughout the iterations; in this case, we say that the method converges to final position if that method is well-tailored for the studied problem. On the other hand, $\vec{r_1}$ (on in the first equation of (12), on the left-hand) and $\vec{r_2}$ (on the second equation of (12), on the right-hand) denote two vectors whose components are randomly generated in the continuous interval $[0, 1]$.

Because the prey's position is generally non-obvious to compute, especially when tackling NP-Hard combinatorial optimization problems containing several local optima, we then simulate such a position according to the best position related to the leader of the pack. Hence, at each iteration of the search process, simulating the positions of the grey wolves hunting may be based upon the three best solutions α, β and δ, and by forcing the other obtained wolves to update their positions according to the position of the best search solution, updated with the following formulas:

$$\vec{D_\lambda} = |\vec{C_\lambda} \cdot \vec{S_\lambda}(t) - \vec{S}(t)|, \tag{13}$$

where λ represents the α, β and δ wolves by affecting their corresponding directions $\vec{D_\alpha}, \vec{D_\beta}$ and $\vec{D_\delta}$ respectively. Differently stated, the three best positions will be updated as follows:

$$\vec{S_1} = \overrightarrow{S^{(\alpha)}} - \vec{A_1} \cdot \vec{D_\alpha}, \quad \vec{S_2} = \overrightarrow{S^{(\beta)}} - \vec{A_2} \cdot \vec{D_\beta} \quad \text{and} \quad \vec{S_3} = \overrightarrow{S^{(\delta)}} - \vec{A_3} \cdot \vec{D_\delta}. \tag{14}$$

Finally, for the next iteration $(t+1)$, each new position related to the current wolf is updated as follows:

$$\vec{S}(t+1) = \frac{\vec{S_1} + \vec{S_2} + \vec{S_3}}{3}. \tag{15}$$

In order to make the paper self-containing, we consider in what follows that $S^{(i)}$ as the solution/configuration of the wolf i in the pack, and $S_1^{(i)}$, $S_2^{(i)}$ and $S_3^{(i)}$ are the computed solutions/configurations when using the three best wolves for computing the new solution/configuration associated to the wolf i.

Algorithm 1 - A standard Grey Wolf Optimizer (GWO)

Input. A starting population $\mathcal{P}$ of size $|\mathcal{P}| > 0$.
Output. $S^{(\alpha)}$: the best position of the final population.

Let $\mathcal{P}$ be the initial population of positions $S^{(i)}$, $i = 1, \ldots, |\mathcal{P}|$.
Evaluate the fitness function z_i related to each wolf, i.e., $z(S^{(i)})$ related to the position $S^{(i)}$, $i = 1, \ldots, |\mathcal{P}|$.
Let $S^{(\alpha)}$, $S^{(\beta)}$ and $S^{(\delta)}$ be the three best positions of $\mathcal{P}$ according to their finesses.
Set $t = 0$ and let t_{max} be a maximum number of iterations.
Initialize a, A and C.
while ($(a \neq 0)$ and $(t < t_{max})$) **do**
  for ($i = 1$ (position of the first wolf of $\mathcal{P}$) to $|\mathcal{P}|$) **do**
    Compute S_1, S_2 and S_3 (according to Eq (14)).
    Evaluate the new position $S_{new}^{(i)}$ of the current wolf (according to Eq (15)).
    if ($z(S_{new}^{(i)}) > z(S^{(i)})$) **then**
      Update $S^{(i)}$ with $S_{new}^{(i)}$.
    end if
  end for
  Update the three best positions of wolves $S^{(\alpha)}$, $S^{(\beta)}$ and $S^{(\delta)}$ according to their fitnesses.
  Set $a = (2 - t \times \frac{2}{t_{max}})$ and update A and C according to Eq (12).
  Increment t.
end while

3.3. A hybrid algorithm for FCMCLP

GWO works with a group of wolves, which can be viewed as a population-based method. Hence, GWO may first start with a randomly generated population $\mathcal{P}$ of wolves (solutions), using a generator trying to cover the search space of the problem concerned for diversification purposes (Section 3.3.1). Second, at each step, regarding the position of the prey, the positions of the wolves should be computed regarding their positions and the positions of the three best wolves in the population (Section 3.3.2). Third, the new population is ranked in decreasing order of their finesses (a maximization problem). Fourth and last, such a process is iterated until matching the stopping criteria. Algorithm 1 describes the main steps of GWO-based population method. In what follows, we mimic such a process for achieving a near-optimal solution for FCMCLP.

3.3.1. A starting population / reference set

Because two versions of the hybrid population-based algorithm are proposed, we then describe how the reference set can be generated. GWO can be viewed as a stochastic method, we then randomly generate a set $\mathcal{P}$ of feasible solutions, where each solution is assigned to a wolf's position.

First version. The starting population $\mathcal{P}$, for the first version of the algorithm, is generated by using the following *Starting Generator Procedure* (SGP), where (In what follows, we consider $\vec{S}^{(i)}$ as the ith wolf/solution):

Algorithm 2 - Starting population Generator

Inputs. Let $\rho = |\mathcal{P}|$ (population size), and m be the number of customers .

Output. A reference set (starting population) for FCMCLP.

Set $\mathcal{P} = \emptyset$ (by setting the population size to $\rho > 0$).
Set $i = 0$ and $nb_{\text{Iter}} = 0$ (the maximum number of iterations is fixed to Max-nb_{Iter}).
repeat
  $i = i + 1$ and set $\overline{S}^{(i)} = \emptyset$.
  repeat
    Set ℓ =rand$(1, m)$.
    if ($\overline{S}^{(i)} \cup \ell$ is feasible) **then**
      update $\overline{S}^{(i)}$=$\overline{S}^{(i)} \cup \ell$.
    else
      close (remove) it from $\overline{S}^{(i)}$.
    end if
  until (providing a complete solution).
  if ($\overline{S}^{(i)} \notin \mathcal{P}$) **then**
    set $\mathcal{P} = \mathcal{P} \cup \overline{S}^{(i)}$
  else
    Decrement(i).
  end if
  Increment(nb_{Iter}).
until ($i = \rho$ or nb_{Iter} =Max-nb_{Iter}).

Second version. The starting population $\mathcal{P}$, for the second version of the algorithm, is generated by cooperating a random generator with the optimal solution of RP$_{\text{FCMCLP}}$, provided with the Simplex method. The reference set is reached by using a modified SGP, namely SGP$_U$, described in what follows:

Algorithm 3 - Starting population Generator using an upper bound

Inputs. $(\bar{x}, \bar{y})$ (RP$_{\text{FCMCLP}}$'s optimal solution), $\rho = |\mathcal{P}|$ (the population size), and m (the number of customers).

Output. A reference set (starting population) for FCMCLP.

1: Set $\mathcal{P} = \emptyset$ (by setting the population size to $\rho > 0$).
2: Set $i = 0$ and $nb_{\text{Iter}} = 0$ (the maximum number of iterations is fixed to Max-nb_{Iter}).
3: `// Generating the first reference solution around the upper bound`
 - Order the vector $\bar{y}$ in decreasing order of their fractional values.
 - Fix the k first $\bar{y}_j$ to 1, $j \in J$, and let $\hat{S}$ be the current partial solution.
4: **repeat**
5:   $i = i + 1$ and set $\overline{S}^{(i)} = \hat{S}$.
6:   **repeat**
7:     Set ℓ =rand$(1, m)$.
8:     **if** ($\overline{S}^{(i)} \cup \ell$ is feasible) **then**
9:       update $\overline{S}^{(i)}$=$\overline{S}^{(i)} \cup \ell$.
10:     **else**
11:       close (remove) it from $\overline{S}^{(i)}$.
12:     **end if**
13:   **until** (providing a complete solution around $\hat{S}$).
14:   **if** ($\overline{S}^{(i)} \notin \mathcal{P}$) **then**
15:     set $\mathcal{P} = \mathcal{P} \cup \overline{S}^{(i)}$
16:   **else**
17:     Decrement(i) .
18:   **end if**
19:   Increment(nb_{Iter}).
20: **until** ($i = \rho$ or nb_{Iter} =Max-nb_{Iter}).

On the one hand, one can observe that the above procedures (Algorithms 2 and 3) try to generate ρ feasible solutions by applying two loops. The external loop may be modified whenever the ρ solutions are not matched (by adding a maximum number of iterations) and so, the population may be completed when calling the core of the method. On the other hand, we can remark the difference between both procedures is related to the use of an upper bound in Algorithms 3 (line 3), especially fixing the k largest fractional values that the Simplex provides; in this case, the Simplex method often favors these variables to optimize the problem instance, hence the strategy is related to fixing these greatest values.

3.3.2. Generating new positions

A *Generational Procedure* (GP) is related to the successive positions induced by overall wolves during the search process. Determining a new position of each wolf is equivalent to combine the current wolf's position and the positions related to the three best positions (solutions) of the population. We recall that Eq. (15) often induces a fractional solution whenever at least one component, at the same rank of each solution, was fixed to one, i.e., the facility is open at this position. Such a formula tries to compute the new wolf's position, which depends on the position of the virtual prey; which is simulated (replaced) to the combination of the three best positions. Herein, we make a cooperation between a series of fractional positions, where each ith integral position (solution) of the population $\mathcal{P}$ may be provided by using the following transformation according to the new achieved position:

1. Let f be the sigmoid function assigned to each element $\overline{S}_j^{(i)}$ of the solution $\overline{S}^{(i)}$, defined as follows:

$$f(\overline{S}_j^{(i)}) = \frac{1}{1 + e^{-\overline{S}_j^{(i)}}} \tag{16}$$

2. Check if $f(\overline{S}_j^{(i)}) \leq \frac{1}{2}$ then set the jth component of the solution $\overline{S}^{(i)}$ to $\lceil \overline{S}_j^{(i)} \rceil$, set it to $\lfloor \overline{S}_j^{(i)} \rfloor$ otherwise.

We note that $\lceil x \rceil$ (resp. $\lfloor x \rfloor$) is the greatest (resp. the lowest) integer value related to x.

Now, one can observe that each component $\overline{S}_j^{(i)}$ may provide a value out of the discrete interval $\{1, \ldots, n\}$ (the greatest(respectively the lowest) bound exceed n (respectively less than 1), where that value reflects the assignment of an opened facility in a certain point p_j. Hence, in order to work with feasibility, we propose the following greedy operator to correct some of these components; that is composed of two steps:

Therefore, the procedure makes sure to provide a new feasible solution, by verifying at each step of including a new opened facility that does not appear in the set $\overline{S}^{(i)}$ (which contains only the current opened facilities). A call for a simple loop is given as follows:

1. Let j be an opened facility in $\overline{S}^{(i)}$.
2. Check the number of occurrences (set it to $occur^{(i)}(j)$) of j in $\overline{S}^{(i)}$.
3. **Do**
   - Replace all the occurrences of j with ℓ, such that ℓ = rand $[1, \ldots, m]$ and $\ell \neq j$.
   While $(occur^{(i)}(j) > 1)$.

3.4. Intensification search

Searching for an improved solution (integral position) is equivalent to fix some stations among the n possible stations. Making some moves between fixed stations is equivalent to fix some of them and to close the rest of the facilities.

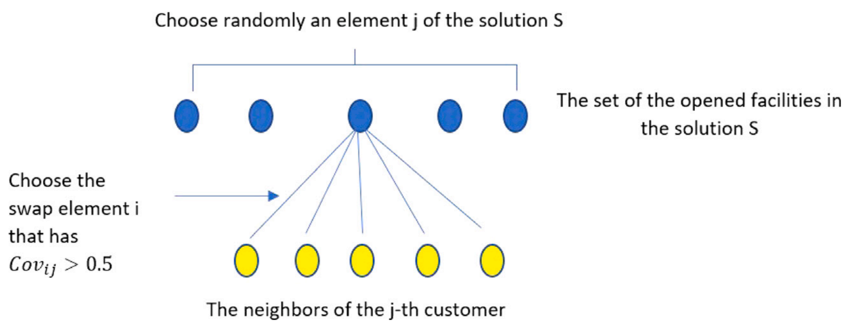

Fig. 3. Simple two permutation for a FCMCLP feasible solution.

Algorithm 4 - Greedy operator

Inputs. Let $\overline{S}^{(i)}$ be the current integral position provided by applying Eq (16).
Output. A feasible solution $\overline{S}^{(i)}$.

Suppose that all components of $\overline{S}^{(i)}$ are unfixed (free).
Let $\delta_k = \sum_{\substack{j=1 \\ j \neq k}}^{n} cov(k, j)$ be the degree of the facility k.
repeat
  Set $\frac{w_k^{(i)}}{\delta_k^{(i)}}$, $k = 1, \dots, n$, in decreasing order, where $\delta_k^{(i)}$ (resp. w_k) denotes the degree (resp. weight) of the facility (resp. the customer) k of the ith configuration.
  Set $\ell = \arg \max_{1 \le k \le n} \left\{ \frac{w_k^{(i)}}{\delta_k^{(i)}} \right\}$.
  Check the feasibility by fixing the facility ℓ (that does not already exist in the solution $\overline{S}^{(i)}$), to close it otherwise.
until providing a complete solution (fixing −close or open−all possible facilities).

3.4.1. A 2-opt operator

Such an operator can be viewed as a simple local search, which is even based upon basic local modifications of the solution at hand. Often, the operator sequentially makes a series of swaps as long as the quality of the induced solution is enhanced. Whenever the solution stagnates, where its objective value remains quasi-unchanged, then we may estimate that the operator converges to a local optimum. Herein, we propose to swap between a current opened facility k of the ith solution $\overline{S}^{(i)}$ with a chosen facility satisfying the following criteria:

$$\ell = \arg \max_{\substack{1 \le r \le n: \\ r = \text{unfixed}}} \left\{ Cov(k, r),\ r \in \{1, \dots, n\}, \right.$$
$$\left. k \neq r,\ k = \text{fixed} \right\}. \tag{17}$$

Of course, for each new opened facility ℓ, the procedure checks the feasibility related to the induced solution.

Fig. 3 illustrates how the 2-opt operator works, regarding a given solution $\overline{S}$. Of course, we can change the condition related to the swapping, between a random element of the solution (which refers to a certain positioned facility) with one of its neighbors. We prefer to do the 2 swap with a new neighbor that has a better coverage degree. The same figure refers to a permutation between the facility j with its neighbor i that has the value related to cov_{ij} greater or equal to $\frac{1}{2}$. Finally, such an operator can be applied for a subgroup of located facilities of the solution $\overline{S}$, and then check the feasibility of the new achieved solution.

3.4.2. Avoid cycling

Often, the 2-opt operator tries to build a series of solutions around other solutions. A subset of solutions may be viewed as a neighborhood according to a given solution. Because a new solution built can be provided by swapping two facilities, one can observe that repeating the same process may lead toward the same local optimum and so, the operator may provide a premature convergence toward local optimum. There are several techniques that can be employed to avoid cycling toward the same solutions. In our study, we considered the 2-opt operator combined with a tabu list in order to mimic the well-known tabu search (Glover, 1989), where the tabu list is characterized by a list of temporary reverse swaps that avoids to return to the solutions already visited.

Because we limit the size of each tabu list, we also used a hashing function, where a list of objective functions is stored during the search process. In this case, a list of Θ bounds (related to the best fitness values) is stored, where each bound may correspond to different solutions. Of course, the smallest the value of Θ is, the more refined results can be achieved, but in this case, it will often be at the expense of the run time. Our limited computational results showed that setting Θ to 20 provided a good compromise between solution quality and the average run time needed by the method.

3.5. Diversification search

Often, the intensification search tries to enhance the solutions at hand. Differently stated, its goal is to provide a series of neighborhoods, issuing from a series of given solutions, which might contain solutions with better quality. Despite some enhancements that can be achieved by such a strategy, and because of the number of achievable solutions with the same objective value, it is interesting to provide a manner able to drive the search process through other unvisited subspaces. In order to simulate such a diversified operator, we propose to add the Drop and Rebuild Operator (DRO) already used in Hifi (2002), where it aims in removing a part of assigned facilities from the current solution. Such a strategy tries to diversify the search process by degrading the quality of the solution at hand hoping to avoid the stagnation in the local optima.

In this case, a first partial (incomplete) solution is built; that is completed by using a part of the procedure described in Section 3.3.2. It can be summarized as follows:

1. Let $\overline{S}^{(i)}$ be the current solution.
2. Drop $\beta\%$ of the fixed facilities belonging to $\overline{S}^{(i)}$.
3. Complete the partial solution by adding the facilities satisfying the best weight per degree, i.e., each added facility ℓ must satisfy

$$\ell = \arg \max_{\substack{1 \le k \le n, \\ k = \text{non-fixed}}} \left\{ \frac{w_k^{(i)}}{\delta_k^{(i)}} \right\}.$$

Algorithm 5 - A Hybrid Population-Based Algorithm (HP-BA) for FCMCLP

Input. An instance of FCMCLP.

Output. The best solution $S^\star$ for FCMCLP.

1: Let $\mathcal{P}$ be the starting reference set of wolfs built by calling either SGP or SGP_U (cf., Section 3.3.1).
2: Apply the generational procedure 4 to $\mathcal{P}$ to ensure the feasibility of overall positions $S^{(i)}$, $i = 1, \ldots, |\mathcal{P}|$ (cf., Section 3.3.2).
3: Set $S^\star$ as the best solution of $\mathcal{P}$ such that $z(S^\star) = \max_{1 \le i \le |\mathcal{P}|} \{z_i(S^{(i)}) \mid S^{(i)} \in \mathcal{P}\}$.
4: Let t_{max} be a maximum number of iterations.
5: **repeat**
6: Let $X^{(\alpha)}$, $X^{(\beta)}$ and $X^{(\delta)}$ be the three best positions of $\mathcal{P}$ according to their finesses.
7: Set $t = 0$ and initialize a, A and C.
8: **while** ($(a \neq 0)$ and $(t < t_{max})$) **do**
9: **for** ($i = 1$ (position if the first wolf of $\mathcal{P}$) to $|\mathcal{P}|$) **do**
10: Let $X^{(i)}$ be the current solution of $\mathcal{P}$.
11: Compute $X_1^{(i)}, X_2^{(i)}$ and $X_3^{(i)}$ (according to Eq (14)), and $X_{new}^{(i)}$ (according to Eq (15)).
12: Check (and improve) the feasibility of the new position $X_{new}^{(i)}$ by calling the 2-opt operator with non-cycling strategy (cf. Sections 3.4.1 and 3.4.2).
13: **if** ($z(X_{new}^{(i)}) > z(X^{(i)})$) **then**
14: Update $X^{(i)}$ with $X_{new}^{(i)}$.
15: **end if**
16: **end for**
17: Set $a = (2 - t \times \dfrac{2}{t_{max}})$ and update A and C according to Eq (12).
18: Update the three best positions of wolves $X^{(\alpha)}$, $X^{(\beta)}$ and $X^{(\delta)}$ according to their fitnesses.
19: Update the best solution $S^\star$ and increment t.
20: **end while**
21: Apply the diversification strategy on the final population $\mathcal{P}$ by calling the drop and rebuild operator DRO (cf., Section 3.5).
22: **until** (satisfying the global stopping criteria)
23: Exit with the best solution $S^\star$.

3.6. An overview of the proposed hybrid population-based algorithm

Algorithm 5 summarizes the main steps of the proposed Hybrid Population-Based Algorithm (HP-BA). It starts (line 1) with a population $\mathcal{P}$ of solutions (related to the wolves) reached by applying one of the constructive procedures discussed in Section 3.3.1. Line 2 ensures the feasibility of the overall solutions of $\mathcal{P}$ by applying the greedy procedure GP (cf., Section 3.3.2). Line 3 stores the best current solution, related the wolf α; that is $S^\star$ whose objective value realizes $z(S^\star) = \max_{1 \le i \le |P|} \{z_i(S^{(i)}) \mid S^{(i)} \in P\}$. Next, the algorithms uses two loops: (i) an *internal loop while* (from line 8 to line 20) and (b) the *global loop repeat* which participates to the diversification of the population for jumping from a region to another non-explored region of the decision space. which applies the GWO for enriching the pack $\mathcal{P}$. Of course, the diversification strategy is that described in Section 3.5, where DRO is applied. Finally, the search process continues till matching the global stopping criteria, where the algorithm exists with the best solution $S^\star$.

4. Computational results

The objective of this part is to assess the behavior of the designed Hybrid Population-Based Algorithm (HP-BA) by comparing its achieved results to those provided by more recent and competitive methods available of the literature and the state-of-the-art Cplex solver. All considered methods were evaluated on the same set of instances extracted from Atta et al. (2022), where each set contains between seven and fifteen values of k, denoting the number of facilities to open, and overall groups are differentiated with the values assigned to m (as discussed below, and displayed in Table 1).

Table 1
Characteristics of the instances.

m	k	$R/f_r/f_d$	S
324	$\{1, 7\}$	600/300/100	4500
402	$\{1, 9\}$	600/300/100	4500
500	$\{1, 13\}$	600/300/100	4500
708	$\{1, 15\}$	600/300/100	4500
818	$\{1, 15\}$	600/300/100	4500
700	$\{1, 12\}$	20/10/5	9500
900	$\{1, 13\}$	20/10/5	9500

The first column of Table 1 represents the different values related to m; which is the number of customers to be served. We note that m belongs to the discrete intervals $\{324, 402, 500, 708, 818, 700, 900\}$. The second column refers to the number of the opened facilities k, as discussed in 1, where R represents the radius of the coverage area of circles of a given facility, $R + f_r$ denotes the fuzzy radius, while f_d is the fuzzy added value to the distance between each customer and installed facility. As underlined in the same table, their values are fixed either to 600/300/100 or 20/10/5. The last column shows the supply capacity of any opened facility in the network, which is either equals to 4500 or 9500.

4.1. Behavior of the first version of HP-BA

4.1.1. Parameter settings

Often, a general purpose heuristic may lead results of variable quality, which is related to a variety of parameters used for tackling large-scale optimization problems. Further, the proposed HP-BA needs three decision parameters: (i) the size of the population, (ii) the number of items to drop for using a re-optimization procedure, and (iii) the criterion used for stopping the search process.

A. effect of the population size

In the preliminary study, the behavior of the first version of HP-BA (without using the upper bound) was tested on the first 28 instances contained in the six groups. Its behavior was evaluated by varying the size of the population, where initially that population was fixed to the minimum size of 5 and doubled for the next version. For each run, the run time limit was fixed between ten and thirty minutes (according to the size of the instance).

Table 2 reports HP-BAs' results, where columns 1 and 2 display the instance information, columns from 3 to 5 show the best upper bounds (percentage coverage, noted $HP\text{-}BA_5$, $HP\text{-}BA_{10}$ and $HP\text{-}BA_{20}$, respectively) achieved by HP-BA according to overall values related to the size of the population, and column 6 to column 8 report HP-BA's run time when varying the population size. Finally, the last line of the table tallies the average lower bound of all tested instances for each version of HP-BA and, the average run time consumed by $HP\text{-}BA_5$, $HP\text{-}BA_{10}$ and $HP\text{-}BA_{20}$, respectively. In what follows, we comment on the results of Table 2:

- For all considered population sizes, all versions of HP-BA are able to match several best bounds achieved by the more recent and the best algorithms available in the literature (extracted from Atta et al., 2022) according to the size of the instance.
- For overall instances, $HP\text{-}BA_{10}$ version (with a population size $Pop_{size} = 10$) reaches a better behavior when comparing its value to those reached by other versions of the algorithm. Indeed, $HP\text{-}BA_{10}$ achieves an average value of 57.71, which realizes a Gap ($HP\text{-}BA_{10}$-$HP\text{-}BA_5$, with a population size $Pop_{size} = 5$) of 0.41 when compared to the average value related to the second best average bounds of HP-BA.

Table 2
Behavior of HP-BA when varying the population size Pop_{size}.

m	*k*	%Cov: solution values			Runtime (min)			Best lit results
		$HP\text{-}BA_5$	$HP\text{-}BA_{10}$	$HP\text{-}BA_{20}$	$HP\text{-}BA_5$	$HP\text{-}BA_{10}$	$HP\text{-}BA_{20}$	
324	1	**37.03**	**37.03**	**37.03**	1	1	1	**37.03**
	2	63.73	**65.33**	62.93	2	2	2	65.14
	5	98.1	**99.07**	95.77	4	10	6	98.82
402	2	56.04	**56.14**	56.04	6	6	7	56.10
	3	**78.47**	**78.47**	77.9	6	8	9	77.25
500	1	**22.83**	**22.83**	**22.83**	1	1	1	22.83
	4	77.8	**80.19**	78.92	20	4	20	79.94
708	1	18.599	**18.6**	18.592	2	4	5	**18.6**
	2	37.14	**37.2**	37.15	6	7	7	**37.2**
700	5	83.67	**83.8**	83.65	30	25	30	83.36
	6	88.46	88.56	**89.09**	11	8	10	87.44
900	2	**26.35**	**26.35**	**26.35**	3	3	3	**26.35**
	4	**56.7**	**56.7**	**56.7**	8	9	10	52.7
Average		57.30	**57.71**	57.15	7.69	**6.77**	8.54	57.14

Table 3
Effect of HP-BA when using DRO.

m	*k*	Variation of the parameter β					
		5%	10%	15%	30%	50%	60%
324	2	64.19	**65.50**	65.20	64.82	65.20	65.20
	5	99.09	96.56	97.62	**99.10**	**99.10**	97.20
	6	98.33	98.64	**99.84**	99.03	**99.84**	99.18
700	2	**34.02**	**34.02**	**34.02**	**34.02**	**34.02**	**34.02**
	4	68.02	68.02	**68.03**	68.02	**68.03**	**68.03**
	7	90.47	90.57	90.57	90.91	**91.32**	90.91
500	2	45.64	45.60	45.61	45.61	**45.66**	45.63
	3	66.01	**66.45**	65.00	66.28	**66.45**	66.28
818	4	54.74	**55.85**	55.62	59.64	59.64	59.64
	5	65.03	65.03	66.30	65.98	**72.00**	68.00
Average		68.55	68.62	68.78	69.34	**70.13**	69.41

Table 4
P-values for both *sign* and *Wilcoxon rank* tests on some instances with the significance level $\alpha = 0.05$.

	μ_1	μ_2	μ_3	μ_4	μ_5
p-value (Sign test)	0.965	0.855	0.637	0.996	0.016
N^+	2	3	4	1	6
N^-	6	5	4	7	0
$N^=$	2	2	2	2	4
p-value (Wilcoxon)	0.869	0.663	0.800	0.987	0.014

- $HP\text{-}BA_{10}$ seems slightly faster than the two other versions of HP-BA; in this case, HP-BA needs an average run time of 6.77 min while $HP\text{-}BA_5$ (resp. $HP\text{-}BA_{20}$ with a population size $Pop_{size} = 20$) consumes 7.69 (resp. 8.54) minutes.

Because we are looking for the version that is able to achieve solutions with high quality (greatest objective values) and with more best bounds, we then fix $Pop_{size} = 10$ for the rest of the computational results.

B. effect of the drop and rebuild operator

In this part, we discuss the effect of the diversification strategy that is based upon the drop and rebuild operator (noted DRO in Section 3.5). We recall that DRO consist in removing a subset of located facilities, which induces a partial solution and, completed it with other facilities by checking the feasibility. Removing located facilities is equivalent to remove $\beta\%$ of these facilities. Hence, the behavior of the method is evaluated on subset of instances, where the run time limit was fixed according to the size of the instance tested (herein, twenty minutes are considered for medium instance and thirty minutes for large-sized ones).

Table 3 reports the bounds achieved by HP-BA when varying the value of β in the discrete interval $\{5\%, 10\%, 15\%, 30\%, 50\%, 60\%\}$. Columns 1 and 2 of the table refer to the instance's information, columns from 3 to 8, under the value assigned to β, display for each instance both the average lower bound achieved by the method over the ten trials. Finally, the last lines summarize the global average bounds achieved by the method. From Table 3, we observe what follows:

- The version with $\beta = 50\%$ (column 7 and the last line, in bold-space) is able to achieve a better global average value of 70.13, when compared to the other versions of the method. With $\beta =$ 50%, the method matches 8 average bounds over the ten instances tested, which represents a percentage of 80%.
- The last version with $\beta = 60\%$ achieves the second best global average bound of 69.41 and it matches 2 better values while the rest of the versions reach a global average value varying from 68.55 (with $\beta = 5\%$) to 69.34 (with $\beta = 30\%$).
- Of course, even the average bound reached by the method with $\beta = 60\%$ remains interesting, but it is still far from the value achieved by the method with $\beta = 50\%$. Indeed, the best version realizes a gap of 0.72, which remains quite consequent for such a problem.

For the rest of the experimental part, and before fixing the value of β, we provide a statistical analysis on the average lower bounds achieved by all versions of the first version of HP-BA, where the parameter β is considered variable as above.

C. statistical analysis

We then apply two well-known statistical measures: (i) the *sign test* which computes both the number of non-negative (>0) and negative (<0) gaps (comparing the achieved gaps related to the difference between two bounds provided by two versions of HP-BA) and, (ii) the *Wilcoxon signed-rank test* statistics that is considered as an alternative study expressing the *p*-value for indicating how one version performs better than another. We do it by setting the following hypothesis: H_0: $HP\text{-}BA_{\beta_1}$ - $HP\text{-}BA_{\beta_2} = \mu$, where $\beta_1 \neq \beta_2$, to express the superiority of the version $HP\text{-}BA_{\beta_1}$ according to $HP\text{-}BA_{\beta_2}$ and, the hypothesis $\overline{H}_0$: the version of $HP\text{-}BA_{\beta_2}$ performs better than the version $HP\text{-}BA_{\beta_1}$ to underline the rejection of the hypothesis H_0. Because we study a maximization problem, we deduce that the greatest the provided bound and the greater the achieved number of better bounds, the better the corresponding version of the method (HP-BA).

Table 4 reports, by varying β_1 and β_2 ($\beta_1 \neq \beta_2$), the study on some instances extracted from the literature, where both *sign test* and *sign rank test* (their corresponding average bounds are those illustrated in Tables 3) are considered. The results reported in the table obtained according to the following study: (i) making a comparative study between two versions according to the values assigned to the parameter β and (ii) returning the best version of the method (i.e., the β inducing a better performance) and continuing the comparison with the untested version. The head of the table (line 0) expresses what follows: $\mu_1 = \mathrm{HP-BA}_{5\%}$ vs $\mathrm{HP-BA}_{10\%}$, $\mu_2 = \mathrm{HP-BA}_{10\%}$ vs $\mathrm{HP-BA}_{15\%}$, $\mu_3 = \mathrm{HP-BA}_{15\%}$ vs $\mathrm{HP-BA}_{30\%}$, $\mu_4 = \mathrm{HP-BA}_{30\%}$ vs $\mathrm{HP-BA}_{50\%}$ and $\mu_5 = \mathrm{HP-BA}_{50\%}$ vs $\mathrm{HP-BA}_{60\%}$. Columns from 2 to 6 show the statistical results of the proposed method HP-BA when varying the parameter β, where the first line of the table (line 1) reports the p-value corresponding to the sign test, the second line (line 2) displays the number of times that the first version of HP-BA with its corresponding β dominates the second version with its corresponding β, the third line (line 3) of the table tallies the number of times that the first version is dominated by the second one, the fourth line (line 4) displays the number of times that both versions are able to match the same average bounds while the last line (line 5) reports the p-value related to the rank sign test. From Table 4, we observe what follows:

- For μ_1 the p-value related to the sign test (resp. Wilcoxon rank-test) is greatest to the significance level $\alpha = 0.05$, indicating that HP-BA$_{10\%}$ performs better than HP-BA$_{5\%}$ (rejecting the hypothesis H_0).
- For μ_2 the p-value related to the sign test (resp. Wilcoxon rank-test) is also greatest to the significance level $\alpha = 0.05$, indicating that HP-BA$_{10\%}$ performs better than HP-BA$_{15\%}$ (rejecting the hypothesis H_0). In this case, HP-BA$_{15\%}$ outperforms HP-BA$_{5\%}$.
- Increasing the value of β improves the quality of the achieved average bounds. In this case, when comparing the three next versions of HP-BA (i.e., $\mathrm{HP-BA}_{30\%}$, $\mathrm{HP-BA}_{50\%}$ and $\mathrm{HP-BA}_{60\%}$), one can observe that $\mathrm{HP-BA}_{50\%}$ performs better than $\mathrm{HP-BA}_{30\%}$ according to Wilcoxon's p-value related to μ_4 (that is equal to 0.987 for $\mathrm{HP-BA}_{30\%}$) and $\mathrm{HP-BA}_{50\%}$ outperforms $\mathrm{HP-BA}_{60\%}$ regarding μ_4 (the Wilcoxon's p-value is closest to 0.014).
- The number of average bounds matched by HP-BA$_{30\%}$, when compared its result to the version HP-BA$_{30\%}$ and HP-BA$_{60\%}$, varies from 6 to 7 for N$^+$ representing $(1 - \mu_4)$ and μ_5. It means that $\mathrm{HP-BA}_{50\%}$ represents the best version of the algorithm.

Hence, because we are looking for the version, which achieve solutions with high quality, with greatest objective values, and with more best average lower bounds, we then decide to fix $\beta = 50$ for the rest of the computational results.

4.2. Performance of the first version of HP-BA vs better available methods

In order to evaluate the behavior of the first version of HP-BA, we compared its provided results to those achieved by the more recent method designed in Atta et al. (2022) (noted MMA) and the state-of-the-art Cplex solver (noted Cplex). Tables 5 and 6 report the solution values achieved by Cplex, MMA and HP-BA on the medium-sized instances (Table 5) and the large-sized instances (Table 6), where their characteristics are detailed in Table 1. For each table, columns 1 and 2 report the instance's information, column 3 displays the best percentage coverage (lower bound) achieved by the Cplex solver, column 4 displays the best percentage coverage provided by MMA (extracted from Atta et al., 2022), while the rest of the columns show the best percentage coverage (Max, column 5) reached by HP-BA and its average percentage coverage (Av, column 6) reached over the ten trials considered.

First, in what follows, we comment on the results of Tables 5 and 6:

- Medium-sized instances:
  1. Column 5 of Table 5 shows the competitiveness of HP-BA when comparing its achieved results to those reached by the Cplex solver. Indeed, its global average percentage coverage is equal to 83.47 while that of Cplex is equal to 83.05. Moreover, HP-BA matches several lower bounds of the literature and dominates the state-of-the-art solver by achieving 18 better bounds, representing a percentage of 62.07% of the best provide bounds.
  2. HP-BA's global average percentage coverage remains also better than that achieved the best method of the literature MMA, where its provided global average bound is equal to 83.41, which remains slightly smaller than that achieved by HP-BA (i.e., 83.47). In this case, HP-BA reaches 12 better lower bounds, matches 10 bounds (achieved by MMA) and fails in 6 occasions.
  3. HP-BA vs both Cplex and MMA: overall instances tested, HP-BA is able to discover 13 new bounds (column 5, in bold-space), it achieves the same bounds in 11 occasions while it fails only in 6 occasions.
- Large-scale instances:
  1. HP-BA remains competitive when comparing its achieved results to those reached by both Cplex solver and the best available values in the literature (MMA), especially for large-scale instances. Indeed, for overall tested instances (of this group), HP-BA's global average percentage coverage is equal to 77.08 while that of MMA is equal to 76.87, as underlined in the last line of Table 6. Further, when comparing the results reached by HP-BA to those provided by Cplex and MMA, one can observe that HP-BA is able to reach 26 (resp. 28) better lower bounds, matches 18 (resp. 24) bounds and fails in 11 (resp. 3) occasions.
  2. Cplex vs other methods: because the Cplex solver fails to solve some large-scale instances (the symbol n/a, column 3), we then reported the global average bounds only on the instances for which the solver was able to solve

Table 5
Performance of HP-BA versus both Cplex and the best method of the literature: medium-sized instances.

m	k	% Coverage		HP-BA	
		Cplex	MMA	Max	Av.
324	1	**37.03**	**37.03**	**37.03**	37.01
	2	65.14	65.14	**65.33**	65.24
	3	86.24	86.24	**86.25**	85.96
	4	92.58	**93.28**	93.10	91.96
	5	98.24	98.82	**99.07**	96.97
	6	99.72	**99.78**	99.54	99.43
	7	99.99	**100.00**	99.96	99.92
402	1	**28.15**	**28.15**	**28.15**	28.15
	2	56.10	56.10	**56.14**	56.03
	3	77.25	77.25	**78.47**	75.48
	4	91.59	91.78	**91.81**	90.90
	5	96.74	96.83	**96.98**	96.15
	6	**99.39**	99.32	**99.39**	99.39
	7	99.47	**99.93**	**99.93**	99.90
	8	99.86	**100.00**	99.73	99.72
	9	99.71	**100.00**	99.77	99.69
500	1	**22.83**	**22.83**	**22.83**	22.83
	2	**45.67**	**45.67**	**45.67**	45.63
	3	66.44	66.44	**66.50**	66.40
	4	80.01	79.94	**80.19**	79.46
	5	87.11	**88.58**	**88.58**	88.58
	6	91.68	92.83	**93.04**	92.87
	7	94.90	95.80	**96.06**	96.05
	8	97.00	98.24	**98.58**	98.57
	9	97.54	**99.26**	99.03	99.03
	10	99.42	**99.72**	**99.72**	99.72
	11	98.69	**99.85**	**99.85**	99.55
	12	99.97	**99.98**	**99.98**	99.98
	13	99.98	**100.00**	**100.00**	99.82
Av.		83.05	83.41	**83.47**	83.12

Table 6
Performance of HP-BA versus both Cplex and the best method of the literature: large-scale instances.

m	k	% Coverage			
				HP-BA	
		Cplex	MMA	Max	Av.
708	1	18.60	18.60	18.60	18.58
	2	37.20	37.20	37.20	37.19
	3	55.80	55.80	55.80	55.80
	4	70.86	70.70	70.70	68.51
	5	80.89	81.33	81.48	81.27
	6	85.28	86.05	86.51	86.51
	7	n/a	88.97	89.11	89.11
	8	n/a	92.13	91.82	91.82
	9	n/a	94.48	94.62	94.62
	10	n/a	97.20	97.66	97.66
	11	n/a	97.83	98.39	98.39
	12	n/a	98.35	98.39	98.39
	13	n/a	99.34	99.44	99.44
	14	n/a	99.59	99.51	99.51
	15	n/a	99.86	99.29	99.29
818	1	15.43	15.43	15.43	15.43
	2	30.86	30.86	30.86	30.86
	3	46.28	46.28	46.28	46.28
	4	59.97	59.97	59.97	59.97
	5	72.42	72.00	72.00	72.00
	6	81.04	81.00	81.00	81.00
	7	84.30	85.61	85.61	85.40
	8	n/a	89.05	89.35	89.35
	9	n/a	92.15	92.26	92.08
	10	n/a	93.35	93.82	93.78
	11	n/a	94.82	95.46	95.46
	12	n/a	95.98	96.40	96.40
	13	n/a	97.11	97.11	97.11
	14	n/a	98.80	98.82	98.03
	15	n/a	99.57	99.57	99.57
700	1	17.01	17.01	17.01	17.01
	2	34.02	34.02	34.02	34.02
	3	51.03	51.03	51.03	51.01
	4	68.04	68.04	68.04	67.99
	5	82.38	83.36	83.80	83.01
	6	85.88	87.44	89.09	87.40
	7	87.06	89.54	90.40	90.30
	8	88.64	91.69	92.73	92.31
	9	89.30	93.21	93.55	92.86
	10	90.23	94.07	94.28	94.07
	11	91.03	94.60	94.61	94.12
	12	91.67	95.36	95.51	94.97
900	1	13.18	13.18	13.18	13.18
	2	26.35	26.35	26.35	26.35
	3	39.53	39.53	39.53	39.53
	4	52.70	52.70	52.70	52.68
	5	65.88	65.88	65.88	65.88
	6	79.05	79.05	79.05	79.05
	7	92.22	92.22	92.22	92.22
	8	97.18	97.54	98.29	98.29
	9	97.43	98.29	98.43	98.42
	10	97.77	98.24	98.94	98.93
	11	98.21	98.55	99.26	99.25
	12	98.28	98.84	99.52	99.49
	13	98.53	98.86	99.33	99.31
	Av.	67.67	68.41	**68.64**	68.43
	Av. Glo.		76.87	**77.08**	76.92

them within the runtime limit. At the end of the table (Line Av.) reports the global average bounds reached by the three tested methods. For this study, we can also observe that HP-BA performs better than both Cplex and MMA.

Fig. 4 shows the variation of the different gaps on the first group of instances (medium-sized instances), where Gap1 = $\%\text{Cov}_{\text{MMA}}$ - $\%\text{Cov}_{\text{Cplex}}$, Gap2 = $\%\text{Cov}_{\text{HP-BA}}$ - $\%\text{Cov}_{\text{Cplex}}$ and, Gap3 = $\%\text{Cov}_{\text{HP-BA}}$ - $\%\text{Cov}_{\text{MMA}}$, respectively. These gaps are computed according to best lower bounds achieved by the three methods and detailed in Table 5.

Table 7
Behavior of HP-BA versus Cplex and MMA: p-values for both *sign* and *Wilcoxon rank* tests on both groups of instances, with the significance level $\alpha = 0.05$.

	μ_1	μ_2	μ_3
p-value (Sign test)	1.000	1.000	1.000
N^+	5	6	9
N^-	27	32	34
$N^=$	35	29	41
p-value (Wilcoxon)	1.000	1.000	1.000

By considering the third gap (Gap3) expressing the difference between the lower bounds provided both MMA et HP-BA on the instances unsolved by the Cplex solver, Fig. 5 illustrates the variation of that gap, which are obtained according to best lower bounds achieved by both MMA and HP-BA on some of the instances (detailed in Table 6).

Finally, Fig. 6 illustrates the variation of the gaps on the second group of instances (large-scale instances), where these gaps are calculated according to the overall best bounds provided by both MMA and HP-BA on overall instances of Table 6.

Second, in order to complete the experimental study, we also made a statistical analysis on the results provided by the three methods. In Table 7, columns from 2 to 4 report the statistical results of HP-BA when compared to the Cplex solver and MMA: line 1 (resp. line 4) tallies the p-value corresponding to the sign test (resp. rank sign test), line 2 displays the number of times that the first algorithm tested dominates HP-BA while line 3 reports the number of times that the first algorithm is dominated by HP-BA. In this case, we set μ_1 =Cplex vs HP-BA, μ_2 =Cplex vs MMA, and μ_3 =MMA vs HP-BA. Of course, we distinguished both cases: (i) the case where the method is able to solve (approximately) all instances (MMA vs HP-BA) and the (ii) the case with the solved instances (Cplex vs MMA, and Cplex vs HP-BA).

From Table 7, we observe what follows:

- p-values related to μ_1, μ_2 and μ_3 are greater than the significance level $\alpha = 0.05$ for both sign and Wilcoxon test. These values indicate that HP-BA performs better than both Cplex and MMA, by rejecting the hypothesis H_0 in all the experiments.
- The number of occasions that HP-BA matches the best average bound is equal to 32 and 34 (line 3, columns 3 and 4) respectively. It also matches the same average bounds in 29 and 41 occasions, respectively, when compared to Cplex and MMA.

Hence, the statistical analysis confirms the competitiveness of HP-BA on overall benchmark instances.

4.3. Effect of the upper bound on the population-based method: HP-BA$_2$

As mentioned in Section 3.3, the study focuses on the design of two versions of HP-BA. The first version studied and analyzed in the first part of the experimentation showed its competitiveness when its results were compared to those obtained by the best methods available in the literature. We study here the behavior of the second version (noted HP-BA$_2$) of the method, which consists in performing a random generation of the starting reference set around an upper bound. From a first solution built according to a fractional variable related to the facilities to be opened (k facilities), the algorithm tries to delimit the positions of the wolves around this bound, in order to avoid the divergence of the wolves.

Table 8 reports the results provided by the second version of HP-BA (noted HP-BA$_2$) and those achieved by the first version HP-BA. In this case, only improved lower bounds achieved by HP-BA$_2$ are reported (for all the rest of the instances both HP-BA and HP-BA$_2$ match the same lower bounds). Columns 1 and 2 report the instance information. Column 3 displays the optimal solution of $\text{RP}_{\text{FCMCLP}}$ (which is an upper bound for FCMCLP). Columns 4 and 5 tally the best lower bounds provided by both HP-BA and HP-BA$_2$. Finally, columns 6 and 7 show

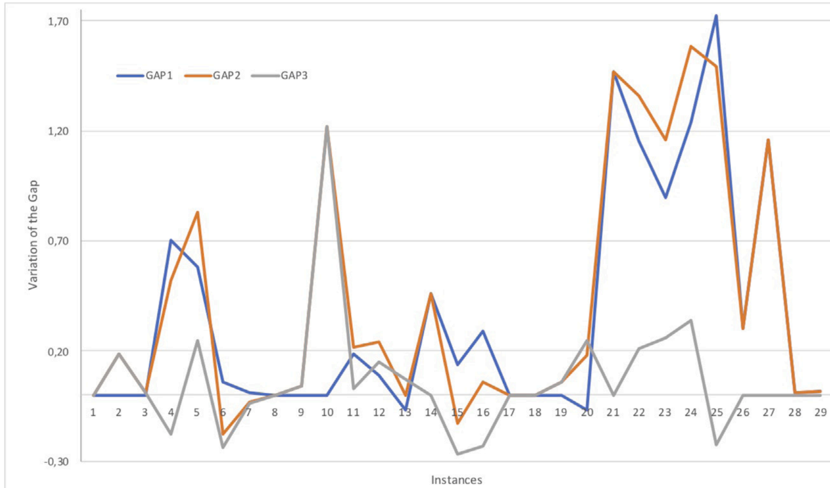

Fig. 4. Variation of the different gaps between the three tested methods (Cplex, MMA and HP-BA) on medium-sized instances.

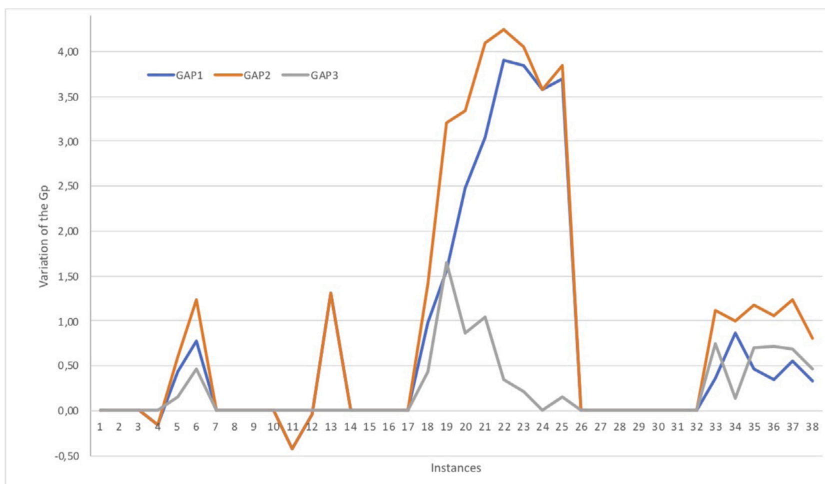

Fig. 5. Variation of the different gaps between the three tested methods (Cplex, MMA and HP-BA) on some large-scale instances.

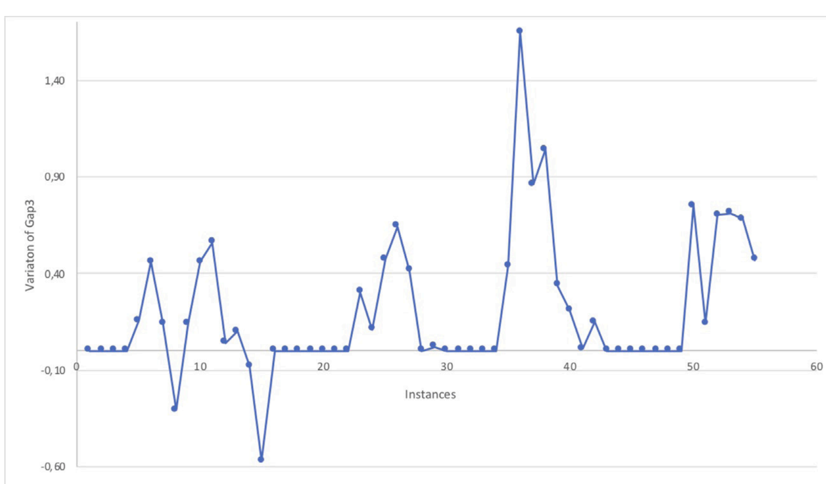

Fig. 6. Variation of the gap representing the difference between both MMA and HP-BA on overall large-scale instances.

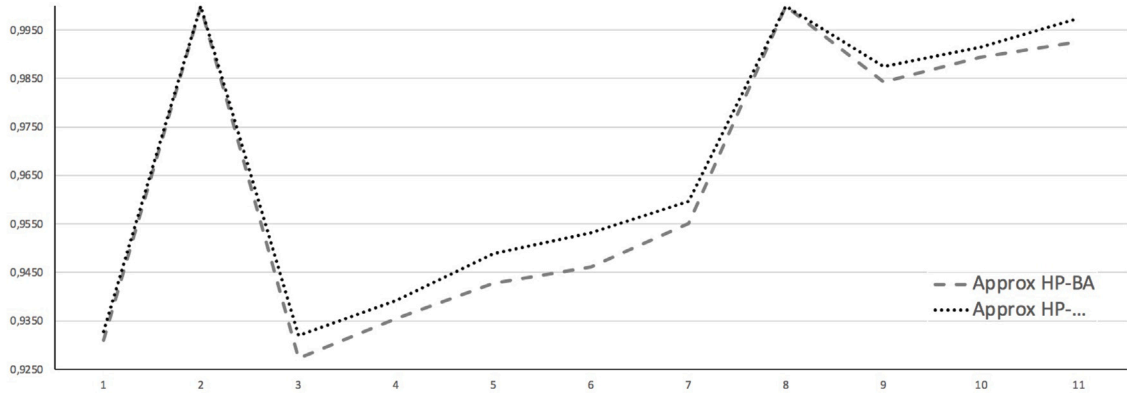

Fig. 7. Variation of the experimental approximation ratio of both HP-BA and HP-BA$_2$ on some instances of the literature.

Table 8
Behavior of the second version of HP-BA (HP-BA$_2$) compared to the first version.

m	k	UB	HP-BA	HP-BA$_2$	A(HP-BA)	A(HP-BA$_2$)
324	4	100	93.1	93.28	0.9310	0.9328
500	2	45.67	45.66	45.67	0.9998	1
700	8	100	92.73	93.1918	0.9273	0.9319
	9	100	93.55	93.92	0.9355	0.9392
	10	100	94.28	94.8845	0.9428	0.9488
	11	100	94.61	95.3122	0.9461	0.9531
	12	100	95.51	95.97	0.9551	0.9597
900	2	26.3521	26.35	26.351	0.9999	1
	9	100	98.431	98.75	0.9843	0.9875
	10	100	98.943	99.15	0.9894	0.9915
	11	100	99.261	99.75	0.9926	0.9975
Average			84.7659	85.1118	0.9640	0.9675

the experimental approximation ratios realized by HP-BA and HP-BA$_2$, respectively. Of course, because the optimal values are unknown (except for some instances, like 500.2 and 900.2, for which the methods match the upper bound), we then, compute each approximation ratio as follows: $\frac{z(\text{HP}-\text{BA})}{UB}$ for HP-BA and $\frac{z(\text{HP}-\text{BA}_2)}{UB}$ for HP-BA$_2$. In what follows, we comment on the results of Table 8:

1. HP-BA$_2$ is able to provide 10 new lower bounds, when compared its results to those achieved by HP-BA (and the best bounds available in the literature). In this case, two optimal solutions are reached by HP-BA$_2$, because it matches both upper bounds (instances 500.2 and 900.2).
2. From the last two columns, one can observe that both HP-BA and HP-BA$_2$ achieve an approximation experimental ratio varying from 0.9273 (instance 700.8) to 0.9999 (instance 900.2) with the first version of the algorithm, while it varies from 0.9319 (instance 700.8) to 1 (instances 500.2 and 900.2) with the second version.

Finally, Fig. 7 illustrates the variation of the experimental approximation ratios of both HP-BA and HP-BA$_2$ on instances for which HP-BA$_2$ dominates the first version of HP-BA.

5. Conclusion

In this paper, the fuzzy capacitated maximal covering location problem was solved with a hybrid population-based algorithm. The method hybridized a discrete grey wolf optimizer with local operators; it combines three main futures. First, a diversified starting population was built by tailoring a random constructive procedure. Second, an exploration search was added in order to augment the quality of the positions of wolves, which characterizes better solutions for the problem; in this case, a special tabu list with a hashing function were also incorporated for avoiding cycling on some visited solutions. Third and last, the drop and rebuild operator was added as an exploration strategy for searching unvisited subspaces around a series of preys' positions related to the three best wolves' positions. Finally, the performance of the hybrid population-based algorithm was evaluated on a set of benchmark instances, where its achieved results were compared to those provided by the state-of-the-art Cplex solver and to the best results extracted from a recent paper in the literature. Several new lower bounds have been discovered.

Through this study on the localization problem, we believe that there are several opportunities for further research involving effective methods capable of improving the proposed method. First, we believe that some preprocessing procedures can be tailored for the studied problem, where tights bounds and valid constraints can be added for reducing the size of the initial instance. In this case, the population-based method can be augmented with adding a fix and solve procedure. Second, the Benders' decomposition can be applied to the studied problem, where the two types of variables can cooperate to provide an iterative method based on the addition of a series of Benders constraints, which can be combined with other valid constraints. Finally, a parallel approach can be considered as another option to solve this problem; in this case, one can imagine partitioning the neighborhoods to design a cooperative parallel method, which can improve the quality of the solutions while reducing the global runtime.

CRediT authorship contribution statement

Méziane Aider: Participated for providing the first version of the paper. **Imene Dey:** Participated to design the hybrid strategy based upon exploration and exploitation operators, Coded the algorithm which is based upon the idea proposed par Pr Mhand Hifi (the supervisor), Tested the final version of the code on a set of benchmark instances extracted from the literature. **Mhand Hifi:** Proposed the main idea related to this work, Supervised all the work, Designed the hybrid strategy based upon exploration and exploitation operators, Provided the first version of the paper, Investigated the sensitivity analysis, Extended the experimental part with a statistical analysis for highlighting the contents of the paper, Provided the final version of the paper.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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